



Student Name: A. Abhishek Reddy

Reg. No.:192424192

Course Code: DSA0104

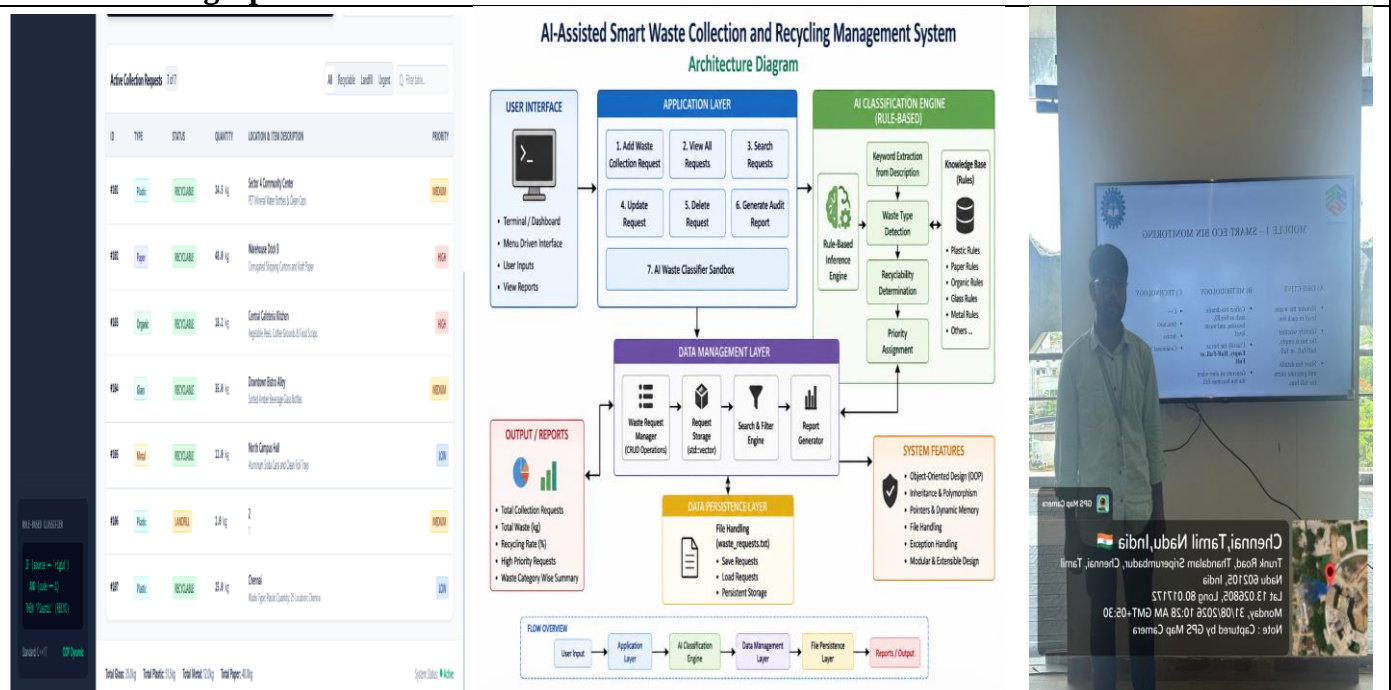
Slot: A

Course Name: OBJECT ORIENTED PROGRAMMING WITH C++

Course Faculty: Dr. PARIMALA KANAGA DEVAN K

Project Title: AI-Assisted Smart Waste Collection and Recycling Management System

Module Photographs:



Project Description

Module 1: Smart Eco Bin Monitoring

The Smart Eco Bin Monitoring module is designed to monitor the waste level in bins in real time. Sensors are used to detect the amount of waste present in the bin, and the collected data is sent to a microcontroller for processing. When the bin reaches a certain fill level, the system generates an alert to indicate that the bin needs to be emptied. This module helps prevent waste overflow, improves waste collection efficiency, reduces unnecessary collection trips, and supports a cleaner and smarter environment. It also helps waste management teams track the status of multiple bins easily. The collected information can be used to plan efficient waste collection routes. Overall, this module provides a simple and effective way to manage waste using smart technology.

Student Signature

Guide Signature